Interpretable Prediction of Reported Older Person's Grant Non-Receipt Among Apparently Eligible Adults in KwaZulu-Natal

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Abstract—This study analysed reported Older Person's Grant (OPG) non-receipt among adults aged 60 or older in KwaZulu-Natal (KZN), using the Living Conditions Survey (LCS) 2014/2015. The study used administrative-burden theory to choose predictors and observable research rules to identify apparently eligible adults. Screening did not establish legal entitlement. Among 1,013 adults who passed with resolved receipt, 67 reported non-receipt: 6.6% unweighted and 7.5% person-weighted. Alternative screening rules gave weighted rates of 7.1%–9.1%. Logistic regression used five predictor groups and household-held-out validation. AUC (area under the receiver operating characteristic curve) was 0.752; average precision (AP) was 0.203. The Brier score was 0.0578, compared with 0.0618 for an intercept-only reference using training-group prevalence. Checks supported reproducibility. The application displays rules, decisions, scores and reports. No participant study tested ease of use. These results do not establish entitlement, explain non-receipt, or demonstrate performance elsewhere.

Keywords—Older Person's Grant, reported non-receipt, administrative burden, Living Conditions Survey, logistic regression, reproducibility.

I. Introduction

The Older Person's Grant (OPG) supports households [1], including during crises [2]. Other work examines receipt and access barriers [3], [4]. This study examines adults who appear eligible but report receiving no OPG.

The analysis separates screening, reported receipt and legal entitlement. Passing the research screen does not mean approval by the South African Social Security Agency [5]. Non-receipt can reflect high income, no application, a pending decision or incomplete information. The analysis must define the screened group before calculating its non-receipt percentage; missing spouse income can change that group.

The study aims to develop and evaluate an interpretable research tool for analysing reported OPG non-receipt among apparently eligible older adults in KwaZulu-Natal using the Living Conditions Survey (LCS) 2014/2015. Four objectives guide the work: document the included population, describe non-receipt using survey weights, test predictions on excluded households, and assess correctness, reproducibility and clarity.

Research questions (RQs) guide the study. RQ1 asks what proportion of screened adults reports non-receipt and how that proportion changes under different screening assumptions. RQ2 asks how well logistic regression separates recipients from non-recipients and estimates chances in excluded households. RQ3 asks whether the tool can repeat the analysis and clearly explain screening decisions and model results. The study combines documented rules, weighted percentages, tested predictions and a local interface to address them.

II. Related Work and Predictor Selection

A. Review Approach, Programme Context and Research Gap

The literature review used thematic narrative synthesis to examine grant access, eligibility measurement and prediction. Searches covered Scopus, Dimensions, IEEE Xplore, SABINET/African Journals Online, Google Scholar and Statistics South Africa/DataFirst. Search concepts included social assistance, eligibility, take-up, non-receipt, exclusion and prediction. Sources were considered for relevance to the research question, population, outcome and methodological contribution. Studies were compared by programme, setting, unit of analysis, outcome, findings and limitations. Differences in these features prevent a pooled non-receipt estimate.

The recorded search summary lists 511 records: 136 duplicates removed, 375 titles/abstracts screened, 222 excluded, 153 full texts assessed, 98 further exclusions and 55 retained sources. These sources cover several grants and populations, rather than 55 OPG studies. The available record documents the search concepts, broad selection criteria and aggregate counts. It does not preserve database-specific query strings, search dates and filters, original exports or record-level screening and exclusion decisions. The full 55-source extraction register is also unavailable. These omissions prevent independent reproduction of the search and verification of each selection decision. The synthesis therefore provides contextual evidence without establishing exhaustive literature coverage.

Table I distinguishes individual receipt [3] from household proportions [4], and this study's provincial resource screen from both. Other-grant studies separate enquiry, application, receipt and uninterrupted payment [6], [7], [8].

Table I. Evidence scope informing the study

Study	Grant / people studied / measure	Contribution and transfer limit
Ralston et al. [3]	Pension; rural older person; receipt	Selection by age in rural Agincourt differs from this study's resource screen.
Godfrey et al. [4]	Grants; Msunduzi household; receipt proportions	Local household findings cannot directly explain individual receipt.
Case et al. [6]; Zembe-Mkabile et al. [7]	Child Support Grant (CSG); enquiry/application and infant receipt	Separates access steps; cannot estimate OPG non-receipt.
Heinrich and Brill [8]; Gibbs et al. [9]; Majola and Maharaj [10]	CSG; interruption and application experiences	Receipt can still require costly effort.
Gazeaud [11]	Targeting assistance using survey measures; resources	Supports checking income assumptions; cannot set South African legal limits.

Gazeaud supports checking income assumptions [11], without quantifying LCS errors or defining South African eligibility. Mabala models another grant for younger applicants [12]; its findings cannot directly predict OPG receipt.

The research gap concerns how these separate strands can support a reproducible, person-level analysis of OPG non-receipt in KwaZulu-Natal. Ralston's age-based rural receipt analysis [3] and Godfrey's household receipt proportions [4] do not directly provide non-receipt estimates after the resource screen used here. Evidence on other grants informs possible access barriers, but does not establish their effects among older OPG applicants. The study addresses this integration gap by documenting the screening denominator, testing how alternative rules change non-receipt estimates, and evaluating an interpretable model on excluded households. Shared rules connect the research command and application so that decisions, probabilities and reports remain inspectable. The specific contribution is a historical workflow that connects transparent screening, person-weighted description, household-held-out prediction and reproducible software. This supports scrutiny of assumptions and results without establishing legal entitlement, causal barriers, participant usability or performance elsewhere.

B. Grant Access and Administrative Burden

Administrative-burden theory distinguishes learning, compliance and psychological costs [13]. Grant studies describe paperwork, repeated visits, interrupted payments and difficult encounters [8], [9], [10], [14]. Child Support Grant research suggests barriers [6], [7], without establishing their effects among older OPG applicants. Digital access can create difficulties [15]. Low non-receipt therefore does not demonstrate easy access.

LCS does not record every legal condition or application experience. The screen checks age, resources, partnership and other-grant rules. It assigns PASS (the person meets the research conditions), FAIL (a known condition is not met) or UNRESOLVED (available information cannot resolve the decision). The analysis then checks receipt separately. Only adults who pass with clear receipt answers enter the prediction study. Unclear or contradictory answers remain unresolved.

C. Predictor Selection and Software Design

Table II separates demographic context from circumstances relevant to learning and compliance costs. These predictors do not directly measure administrative burden or establish why a person reports non-receipt.

Table II. Administrative-burden theory and predictor selection

Predictor	Theoretical role	LCS coding and limit
Age	Demographic context: receipt differs by age [3]	Per decade above 60; health and application history unmeasured.
Sex	Demographic context: receipt differs by sex [3], [4]	Female versus male; social roles and discrimination unmeasured.
Schooling	Learning costs: schooling may help people find and understand grant information [10], [13]	Below Grade 7 versus Grade 7 or above; awareness and literacy unmeasured.
Settlement	Compliance costs: place may shape service access [6], [14], [15]	Urban informal and traditional/rural formal, each versus urban formal; office capacity unmeasured.
Office distance	Compliance costs: distance may add travel effort [10], [14]	At least 5 km/unavailable versus nearer than 5 km; visits and travel costs unmeasured.

Psychological costs are unmeasured. Settlement contributes two contrasts, giving five predictor groups and six slopes.

Few events motivate assessment of precision, calibration and ranking [16], [17]. Sample size and internal validation limit generalisation [18], [19]. Reporting follows TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) [20]. Software checks assess correctness and repeatability; usability requires participant evidence.

D. Conceptual Framework

Reported OPG non-receipt among apparently eligible older adults motivates an administrative-burden perspective [13], [14]. Learning, compliance and psychological costs may obstruct access; high receipt does not establish low burden.

Figure 1 separates theoretical constructs from LCS measurements. Schooling may reflect information resources; office distance and settlement relate to potential compliance effort. Age and sex provide demographic context. These proxies do not directly measure burden. Screening defines the denominator because age alone is insufficient and LCS cannot establish legal entitlement. Resource, partnership and other-grant rules define the population; incomplete information remains unresolved. Reported receipt defines a separate outcome.

Person-weighted estimates describe non-receipt. Logistic regression describes associations; household-held-out

validation tests predictions. Neither establishes causal mechanisms. The artefact implements the design through shared rules, separate receipt classifications, saved-model scoring and inspectable reports. Upload scores remain separate from held-out evaluation and never change screening or receipt.

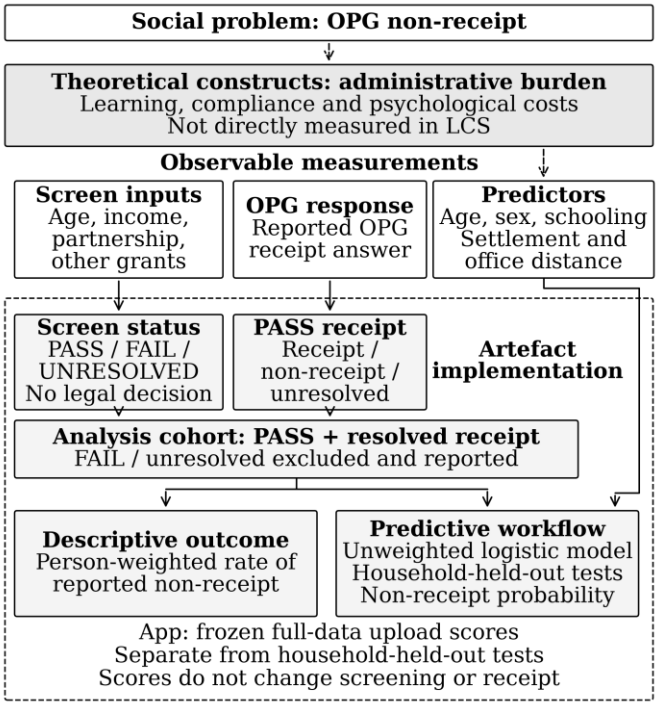


Fig. 1. Conceptual framework. Theory informs predictor selection; observable screening defines the population for separate descriptive and predictive analyses. The dashed boundary identifies the artefact's implementation scope. Arrows show conceptual rationale or analytical dependencies, not causal effects.

III. Methods

A. Data and study population

The analysis linked the LCS 2014/2015 person, household and person-income files through survey identifiers. Checks covered duplicates, unmatched records and row counts. The questionnaire, metadata and COICOP (Classification of Individual Consumption According to Purpose) codes guided interpretation [21], [22], [23]. The study covered sampled KwaZulu-Natal residents aged 60 or older in completed years. The survey excludes older adults living in institutions. Screening used the rules for each interview month. Before releasing the data, Statistics South Africa used hot-deck imputation for age and sex. The released files do not identify these records. The model handled missing predictors separately.

Table III shows each field's role. Summing income before linkage avoided duplicate person records.

Table III. How each data field enters the analysis

Data concept	LCS measurement	Use and restriction
Person/household identifiers	Household and person codes	Link files, identify spouses and keep households together for model checks; excluded from model inputs and public reports.
Age/province	Completed years; province code	Select KZN adults aged 60+; model also uses age.
Partnership	Marital/resident-partner answers; spouse identifier	Select resource rule and link spouses; excluded from primary model.
Resources	Yearly income items and source schedule	Screen using estimated private resources; no allocation of household income.
OPG receipt	General grant/OPG answers and skipped questions	Define outcome after PASS; exclude receipt and derived labels from predictors.
Other/prior grants	Own grants; preceding-year OPG income	Check incompatible grants and reporting periods; excluded from predictors.
Schooling/sex/place	Education, sex, settlement, office distance	Primary predictors; fill missing values using training data only.
Weight/design	Person weight; derived survey group	Describe population and test alternatives; excluded from primary predictors.

Released documentation supports these field uses [21], [22], [23]. Filling missing model predictors never changes screening fields or receipt answers.

B. Screening rules and receipt outcome

The screen summed already annualised, nominal person-income amounts, excluding social-assistance income. It did not allocate household income. A missing applicant or linked-spouse amount remained unresolved unless the no-income response and absent schedule supported zero. Assets, deemed income and permitted deductions were outside this research proxy.

GN 211 set the standard monthly maximum at R1,350 from 1 April 2014 [24]; GN 277 raised it to R1,410 from 1 April 2015 [25]. Government Notice R211, a separate amendment in the same gazette, substituted $D = 1.6A - 0.4B$ from 1 April 2014 in Schedule paragraph 2 [26]. With annual standard maximum A and minimum D = R1,200 (Annexure A(5)) [27], $B = (1.6A - 1{,}200)/0.4$. This yields annual single/couple research ceilings of R61,800/R123,600 for October 2014–March 2015 and R64,680/R129,360 for April–October 2015. Half-joint income doubles couple limits; equality passed. The screen applied these boundaries to its private-income proxy. Both notices add R20 for age 75+; the primary screen retained the standard maximum across ages, while S1c tested the supplement. S1 also tested period and payment-boundary alternatives.

The 2008 regulations define spouses (regulation 1), include spouse income regardless of property regime (18(2)) and use half-joint annual income (19(1)). They allow Agency-approved exclusion after continuous desertion for at least three months and unwillingness to support (18(3)) [27]. The July 2015 amendment left these OPG means provisions unchanged [28]. Survey responses established neither legal spouse status nor desertion conditions. Never-married, widowed and divorced adults used single ceilings. For married adults, research rules required resolved applicant/spouse incomes and reciprocal pointers linking a resident spouse; sums applied regardless of spouse age. Nonresident spouse income was unavailable; separated respondents lacked spouse details [23]. Cohabitation, unspecified status and missing/inconsistent links remained unresolved, without assumed single status or zero partner income. S2 tested alternatives.

An adult's reported disability or war-veterans grant caused FAIL. Caregiver child grants and grant-in-aid neither excluded the adult nor defined OPG receipt. A known failure took priority over unresolved information; the screen retained all reasons.

For adults who passed, the software interpreted general-grant and OPG answers together, including structural skips (questions omitted because earlier answers make them inapplicable). A consistent “no grant” answer can establish non-receipt when the OPG follow-up does not apply. Contradictory or unexplained answers remain unresolved and exclude the person from modelling. The analysis flagged disagreements between current receipt and preceding-year OPG income without changing the outcome because the periods differ. Rule version opg-kzn-primary-v1 records these decisions. The denominator contains adults who pass and have a clear receipt answer.

C. Survey-weighted description and uncertainty

RQ1 used both the ordinary count proportion and the person-weighted proportion: $p = \Sigma(w_i Y_i)/\Sigma w_i$. Here, w_i is a person's weight and Y_i equals 1 for non-receipt and 0 for receipt. Group comparisons describe differences; they do not establish causes.

The released files lack strata. Derived primary sampling units (PSUs) supported an estimate of sampling uncertainty. The calculation allows a group to be sampled more than once. It cannot reproduce the complete survey design. Adults with unresolved screening information remain outside the proportion. Receipt bounds (the lowest and highest rates if all unclear PASS receipt answers mean receipt or all mean non-receipt) tested the effect of unclear receipt answers.

D. Model specification and validation

The unweighted logistic model estimated probability of reported non-receipt. The analysis did not rebalance recipients and non-recipients, automatically select variables or choose a cutoff. The model describes associations after accounting for other predictors. Age enters per decade above 60. Comparison categories are male, Grade 7 or above, urban formal and an office nearer than 5 km. Five predictor groups require six slopes and an intercept, because settlement requires two comparisons.

Rural formal settlement had 31 adults and one non-recipient. One seven-slope training fold then contained 30 rural-formal adults and no event; its coefficient was about −21.88 and did not converge. Four of five folds converged. Decision D-052 combined rural formal with traditional settlement after that outcome pattern was known, so the retained cross-validation can be optimistic and does not show that the settings are equivalent. The seven-slope out-of-fold AUC was 0.739 and its calibration slope 0.673, against 0.752 and 0.836 after the merge. Because one fit failed, this comparison is diagnostic only. Ridge logistic regression was then applied to both codings on the same household folds. The penalty was chosen inside each training set from a fixed grid. Every fold selected the same penalty, and all ten fits converged. Held-out AUC was 0.756 for the merged coding and 0.750 for the separate coding (Supplementary Table S2). The ordinary merged model remains the reference. This comparison does not make Decision D-052 outcome-independent. Income, receipt, derived labels and marital status do not enter the primary model.

Validation used five household-grouped folds. Sorted households were labelled by whether any member reported non-receipt, then stratified so every adult from a household stayed in one fold. Seed 201415 fixes this division. The 1,013 adults come from 827 households: 182 contain more than one modelled adult, and 6 contain both receipt and non-receipt. Each adult received one out-of-fold (OOF) probability from a model trained without that household. The intercept-only reference uses the training group's non-receipt rate.

After interpreting skipped questions, each training fold supplied the fill-in for its held-out adults. Schooling contrasted below Grade 7 with Grade 7 or above, and office distance contrasted at least 5 km or unavailable with nearer than 5 km. Other codes stayed missing. The training fold filled missing age, in decades above 60, with its median, and a missing category with its most frequent code, or the smaller code in a tie. Eligibility, receipt and weights were not filled or modelled. Three recipients lacked usable schooling; 14 recipients and four non-recipients lacked usable office distance. Age, sex and settlement were already complete.

Pooled OOF probabilities supplied average precision, AUC, Brier score, calibration slope and calibration-in-the-large. Calibration-in-the-large is an intercept correction with its slope fixed at 1; the ideal values are 0 and 1. The 95% intervals are 2.5 and 97.5 percentiles from 1,000 with-replacement household samples of those fixed predictions, using the same seed. Every selected household's adults were retained. All 1,000 samples contained both outcome classes. These intervals condition on seed 201415, its preprocessing and fits, and Decision D-052. They do not repeat the split, imputation or estimation, and they are not survey-design confidence intervals. A separate sensitivity repeated the same household-grouped procedure 20 times, using seeds 201415–201434, with training-only imputation and refitting. Decision D-052 stayed fixed.

E. Alternative assumptions

S1 changes thresholds, S2 partnership assumptions, S3 rules about other grants, S4 income definition, S5 receipt period and S6 unresolved-receipt bounds. Each records the denominator, non-recipients, ordinary/person-weighted rates and internally validated model performance. S7 uses complete cases. S8 replaces welfare access with a married indicator. S9 uses

frequency weights (treating person weights as repeated-record contributions), keeping unweighted missing-value treatment and evaluation. This is not a full survey-design regression. S10 weights primary OOF measures without refitting. S11 groups validation folds by derived PSU. These results remain separate from the primary analysis.

IV. Software and Evaluation

The command python analysis/run_study.py links survey files, screens adults and calculates results. Run records retain settings, versions, seed, folds and checksums. The saved model stores coefficients and missing-value settings in JSON, without executable program code. Version opg-kzn-six-slope-v1 uses the model trained on all included adults. Uploads do not retrain it.

The local Streamlit app has five sections: study context, upload/validation, eligibility/receipt, prediction/model and export/reproducibility. It accepts historical opg-screen-v1 CSV files, including household members needed to check spouse resources. Validation rejects incorrect schema, study context, duplicate identifiers, categories and numeric values. Limits are 20 MiB and 50,000 rows. Uploads stay in memory during the current session. Reports use generated row labels and omit submitted identifiers, raw income and household links. Users can filter views and exports; summary rates retain all uploaded adults who pass with resolved receipt.

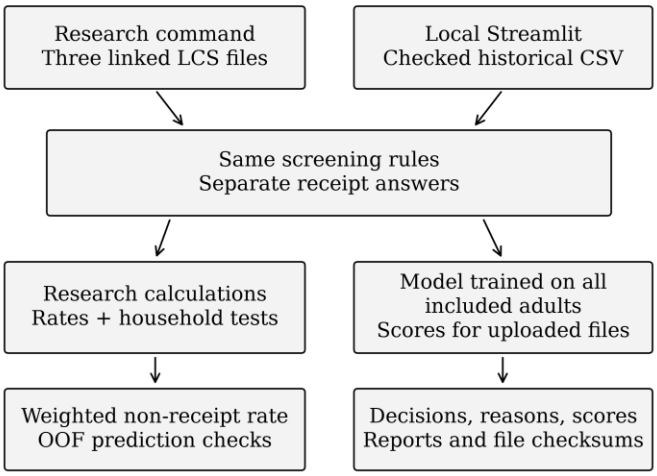


Fig. 2. Research and app workflows share screening rules. Household-held-out predictions remain separate from app scores calculated using the model trained on all included adults.

The app reuses the screening rules and saved model settings. It scores only supported adults who pass with resolved receipt. Saved training statistics fill partly missing predictors; records missing every categorical predictor remain unscored. Integrity checks link model and evaluation files to the analysis and predictor code. The interface labels historical OOF results separately from uploaded scores; uploaded records may overlap training data. CSV, JSON and ZIP reports contain classifications, reasons, supported probabilities, versions and an audit manifest (settings and file checksums). The downloadable package uses invented data and excludes survey microdata.

Software checks compared screening decisions with expected results and tested excluded predictors, links, repeated runs, reports and saved predictions. Further checks compared the CLI (command-line interface) with the UI (user interface), including invalid inputs and outdated results. The owner's Windows walkthrough demonstrated local functionality. A task sheet supports future testing with three to five consenting participants; no participant study has taken place.

V. Results

A. Study group and reported non-receipt

The survey included 15,388 people in 3,686 KwaZulu-Natal households. Table IV shows the selection steps. Among 117 failures, 93 exceeded the income ceiling and 25 reported a disability grant; one met both conditions. None reported a war-veterans grant. Among 187 unresolved adults, 105 were married, mostly with a non-resident spouse, and 63 reported living together. Unresolved adults outnumbered non-recipients.

Table IV. KwaZulu-Natal study group at each screening step

Step	People	Households
KwaZulu-Natal respondents	15,388	3,686
Aged 60 or older	1,317	1,068
Failed observable rule	117	98
Eligibility unresolved	187	168
PASS and resolved receipt	1,013	827
Reported receipt	946	773
Reported non-receipt	67	60

Each row counts distinct households. A household can appear in several rows.

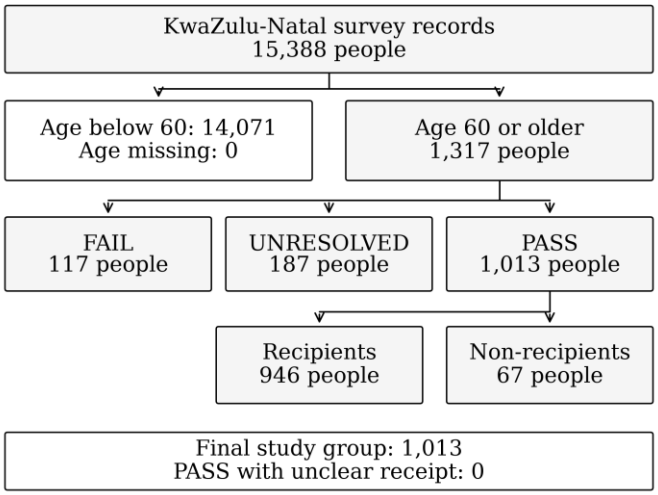


Fig. 3. FAIL, UNRESOLVED and PASS groups do not overlap. All 1,013 PASS adults resolve receipt and enter the final denominator.

Of these 1,013 adults, 67 reported non-receipt: 6.61% without weights and 7.49% with person weights. The calculation used 345 derived PSUs and gave a standard error of 1.3 percentage points. A bell-shaped sampling approximation gave a 95% confidence interval of 4.9%–10.0%. Missing strata limit this approximation. All 1,013 PASS answers agreed with preceding-year OPG income, so those checks changed nothing. Table V separates unweighted sample summaries from person-weighted summaries before missing values are filled. Percentages describe receipt-group composition, not non-receipt rates or predictions.

Table V. Characteristics by reported OPG receipt, before imputation

Characteristic	Recipients, N = 946	Non-recipients, N = 67
Age, mean years	70.4 unw.; 69.8 wtd	67.3 unw.; 68.4 wtd
Female	686 (72.5%) unw.; 70.4% wtd	33 (49.3%) unw.; 53.4% wtd
Grade 7 or above	290 (30.7%) unw.; 33.0% wtd	49 (73.1%) unw.; 80.8% wtd
Schooling missing	3 (0.3%) unw.; 0.4% wtd	0 (0.0%) unw.; 0.0% wtd
Urban formal settlement	284 (30.0%) unw.; 36.2% wtd	38 (56.7%) unw.; 66.2% wtd
Office ≥5 km or unavailable	717 (75.8%) unw.; 72.0% wtd	34 (50.7%) unw.; 40.3% wtd
Office distance missing	14 (1.5%) unw.; 1.8% wtd	4 (6.0%) unw.; 7.2% wtd

unw. is the unweighted count and percentage, or the unweighted mean age. wtd is the person-weighted percentage or mean. Missing predictors remain in the denominators. Supplementary Table S4 prints the unweighted and weighted values in separate columns. These summaries have no uncertainty intervals.

B. Screening sensitivity

Table VI keeps scenarios whose counts differ from the primary screen. Changed cohorts were refitted. Unchanged cohorts reused the primary predictions, so matching performance is not an independent fit. Supplementary Table S1 defines every rule.

Table VI. Changed screening rules and OOF performance

ID	N/events	Unw. %	Wtd. %	AUC	Brier	Slope
S0	1,013/67	6.61	7.49	.752	.0578	.836
S1b,d	1,015/69	6.80	7.68	.763	.0584	.875
S2a	1,072/71	6.62	7.42	.749	.0588	.779
S2b	1,073/69	6.43	7.17	.756	.0564	.841
S2c	1,021/67	6.56	7.43	.753	.0571	.839
S2d	1,081/69	6.38	7.12	.751	.0565	.808
S3	1,032/82	7.95	9.06	.736	.0682	.864
S4b	1,007/67	6.65	7.55	.755	.0576	.854
Same *	1,013/67	6.61	7.49	.752	.0578	.836

Events are reported non-recipients. Performance uses no weights. S1b uses the 2015 ceiling at every interview. S1d uses the income level at which the formula pays nothing; it admitted the same two adults as S1b. S2a/S2b screen living-together adults as couples/single. S2c screens separated adults as single; S2d screens both as single. S3 retains disability-grant recipients and adds no war-veterans recipient. S4b keeps social-assistance income in the proxy.

*S1a, S1c, S4a, S4c, S5 and S6a/S6b keep the S0 counts and reuse its predictions. Identical counts do not make the rules equivalent. No PASS adult has unresolved receipt for S6a/S6b to bound. Definitions are in Supplementary Table S1.

C. Held-out prediction

All five folds converged. Figure 4 uses five similarly sized bins. The highest bin had average prediction 0.188 and 35/203 non-recipients (17.2%); the lowest had average 0.0107 and 3/206 non-recipients (1.5%).

Table VII. Primary held-out performance and uncertainty

Metric	Logistic: estimate [95% interval]	Intercept-only: estimate [95% interval]
AUC	.752 [.687–.814]	—
Average precision	.203 [.139–.292]	—
Brier score	.0578 [.0448–.0706]	.0618 [.0467–.0758]
Calibration-in-the-large	.020 [−.316–.273]	.0001 [—]
Calibration slope	.836 [.591–1.133]	—

Estimates pool unweighted OOF predictions: 1,013 people/67 events. The intervals use percentile bounds from 1,000 household resamples of fixed predictions. All reported intervals have zero invalid resamples. Resampling did not refit models or missing-value settings. The intercept-only reference uses the training group's non-receipt rate. Dashes mean unavailable measures/intervals, rather than zero or chance performance.

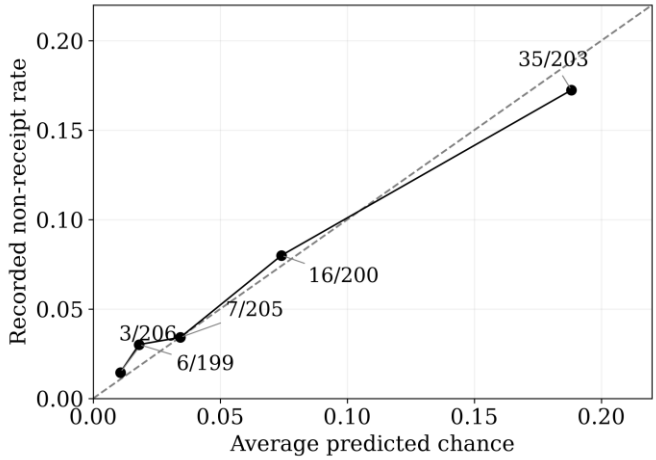


Fig. 4. Saved group summaries show unweighted OOF calibration. Labels give events/people; the diagonal shows exact agreement between average predictions and observed rates. Plot range: 0–0.22.

The odds ratio for women versus men was 0.39 (Wald interval 0.23–0.66). Schooling below Grade 7 versus Grade 7 or above gave 0.24 (0.13–0.43). These full-sample associations are not held-out performance or causes. Only 67 events supported six slopes. Across the 100 refitted training folds, the female and below-Grade-7 coefficients stayed negative, while the urban-informal coefficient changed sign. Age, settlement and welfare-distance intervals included 1. That is limited precision, not evidence of no association, and those intervals omit the settlement-coding decision. The 20 repeated splits gave 100 converged fits. AUC ranged from 0.733 to 0.773 (median 0.753) and the calibration slope from 0.738 to 0.943 (median 0.837). Supplementary Table S3 gives the median, interquartile range and range for every performance measure, and the coefficient ranges. That spread is a sensitivity summary for the fixed coding, not a confidence interval.

Table VIII. Model/validation sensitivities

Scenario	N/events	Unw.%	AUC	Brier	Slope
S0 primary	1,013/67	6.61	.752	.0578	.836
S7 complete	992/63	6.35	.758	.0556	.819
S8 married	1,013/67	6.61	.755	.0578	.846
S9 wtd. fit	1,013/67	6.61	.750	.0583	.718
S10 wtd. measures	1,013/67	6.61	.763	.0641	.935
S11 PSU folds	1,013/67	6.61	.758	.0575	.860

The non-receipt-rate column uses no weights; S8–S11 keep the weighted non-receipt rate at 7.49%. S9 uses frequency weights when fitting, but no weights when filling missing values or checking predictions. S10 weights the primary OOF predictions without refitting. S8 replaces welfare access and keeps six slopes. S7 removes 21 people/four events. S11 changes fold grouping but keeps the study group.

D. Software verification

The 26 checks in Table IX included 22 synthetic screening cases (invented records with specified expected decisions), two income-schedule checks, one forbidden-predictor check and one run trace. Boundary, spouse, grant and receipt decisions matched their specifications. Two research runs produced 14 files with exactly matching contents and retained the study-group checksum.

Table IX. Completed verification and its scope

Evidence	Observed result	What it supports
Original checks	26/26, including 22 screen cases	Check rules, schedules, excluded predictors and run tracing.
Complete automated suite	220 pass; 0 fail/skip	Check software, app state and exports; includes 18 Streamlit interface tests.
App–command agreement	15,388/15,388 agree; 1,013 scored	Both paths select the same people and give the same decisions, receipt, reasons and outcome.
Reloaded model scores	Maximum score difference 5.55×10^{-17}	Reloading preserves saved-model scores; this does not assess OOF performance.
Owner Windows exports	50/50 checks of six files	Check synthetic local use and report consistency.
Participant usability	Not conducted	No representative task-time, error or ease evidence.

The owner walkthrough used synthetic records. It checks exports, not representative task times or ease.

VI. Discussion

A. Scope of measured non-receipt

RQ1 measures non-receipt among historical KwaZulu-Natal adults who pass the screen and resolve receipt: 67/1,013, or 6.61% unweighted and 7.49% person-weighted. This does not measure unlawful denial or non-receipt among everyone legally entitled. The 187 unresolved adults outnumbered the 67 events. Excluding them changes the population described, regardless of sampling uncertainty. Stable estimates cannot reveal their receipt.

S2d added 68 people but only two events, reducing weighted non-receipt to 7.12%. Partnership rules mainly changed the denominator. S3 retained adults with incompatible grants, added 19 people/15 events and raised non-receipt to 9.06%, the largest tested change. Similar AUC cannot justify screening rules or establish equivalent populations. Ralston studies rural age-eligible people [3]; Godfrey studies households' grants [4]. Different units, programmes, screens and places prevent direct comparison of their percentages. This study adds a provincial estimate with an explicit resource screen. High receipt can involve access costs [13], [14]; it cannot prove low burden.

B. Predictive performance and interpretation

RQ2 assesses four properties. AUC 0.752 describes ranking held-out non-recipients above recipients, not classification accuracy. AP 0.203 summarises precision–recall ranking across cutoffs against a 6.61% event rate, not precision at one cutoff [16]. Brier fell from 0.0618 to 0.0578: a modest 0.0040 reduction in squared probability error. An intercept correction, with calibration slope fixed at 1, checks average bias. The ideal correction is zero. The estimate 0.020 and interval spanning zero do not prove perfect agreement. Slope 0.836 suggests predictions may be too extreme, but its 0.591–1.133 interval includes the ideal 1 [17]. Only 67 events informed these checks [18]. Most adults in even the highest-risk bin reported receipt.

Table VIII separates sample, predictor and evaluation changes. S7 removed 21 people/four events (992/63). Its lower Brier concerns a different sample and cannot establish whether filling missing values is better. S8 substituted marriage for welfare access and performed similarly. S9 weighted fitting but not evaluation; it does not model the full survey design. S10 reweighted the same primary OOF predictions without retraining, raising Brier to 0.0641. S11 grouped folds by derived PSU and also performed similarly. S8–S11 retained 1,013 people/67 events. These checks expose internal assumptions; they do not establish model superiority or reliability elsewhere [19], [20].

C. Reproducibility and usability

RQ3 has six evidence layers. Synthetic cases check rule decisions; matching file contents check repeatability; UI–CLI agreement checks software paths; JSON reload checks saved-model predictions. The owner walkthrough checks a synthetic workflow; participant testing would assess human usability. Table IX supports the first five within tested conditions. Automated agreement cannot establish independent user ease, acceptance or field success. Saved sources and run/validation reports document versions, checksums and test conditions.

The app separates historical OOF results from uploaded scores; uploads may overlap training data. Matching saved scores checks reloading, not new predictive accuracy. The software exposes its computations, but representative user clarity remains untested. The task sheet supports future testing.

D. Transfer to other settings

This study analysed screened KwaZulu-Natal adults in LCS 2014/2015. Provinces and periods differ in settlement, services,

policy and reporting. Application, award and payment records measure different events from reported receipt and can contain linking errors. The study did not test the model on an independent province, period or administrative sample. Transfer requires comparable populations/outcomes, justified period-specific rules and independent ranking/calibration checks [19], [20]. Making the app available cannot establish suitability for today's applicants.

E. Synthesis of the Conceptual Framework

Returning to Figure 1, screening sensitivity supported an explicit denominator, and sex differences supported demographic comparison without confirming burden mechanisms [3], [4]. Lower schooling accompanied receipt, contrary to a simple expectation that fewer information resources increase non-receipt [13]. Screening selection or other resources could account for this; neither explanation was tested. Office-distance and settlement intervals included 1, leaving compliance-cost expectations unresolved. Psychological costs remained untested because LCS did not measure them. Proxy associations do not directly test the theory. The artefact contributes a reproducible implementation of these bounded measurement choices. Separate, traceable rules, outcomes and predictions support inspection despite measurement limits, without establishing causes, legal entitlement or participant usability.

VII. Limitations and Ethics

The survey cannot establish legal eligibility: its resource proxy omits assets, deemed income and deductions. Respondents may misreport income and receipt. Cross-sectional data cannot establish the order of events. Few events, unresolved screening, missing strata and resampling fixed predictions without refitting limit inference. Household sampling excludes institutionalised older adults. Participant usability and performance elsewhere remain untested.

Survey access conditions and disclosure requirements apply. Interface demonstrations and tests use synthetic records. The downloadable package excludes survey records. Displays and reports omit submitted identifiers and raw income. These controls do not establish disclosure review of every historical repository file. The project records contain no DUT (Durban University of Technology) ethics determination. The paper therefore claims no approval. The app explains measurement choices; it neither decides grants nor ranks named applicants.

VIII. Conclusion

Historical screened non-receipt was 6.6% unweighted and 7.5% person-weighted; alternatives gave 7.1%–9.1%. Rules about other grants changed results more than the ceiling, and unresolved cases made the denominator central. Five predictor groups/six slopes gave AUC 0.752, AP 0.203 and Brier 0.0578: moderate internal ranking with slightly less probability error. The software reproduces screening, receipt categories and scores for inspection under tested conditions. The study provides a historical benchmark. It does not establish human usability, causes, legal entitlement or performance elsewhere. Future work should validate the frozen model in independent provincial or later-period data, using comparable screening and receipt definitions, and should assess discrimination, calibration and probability error before any updating. Intended users should also complete synthetic-data tasks that record task success, errors, completion time and perceived ease of use.

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